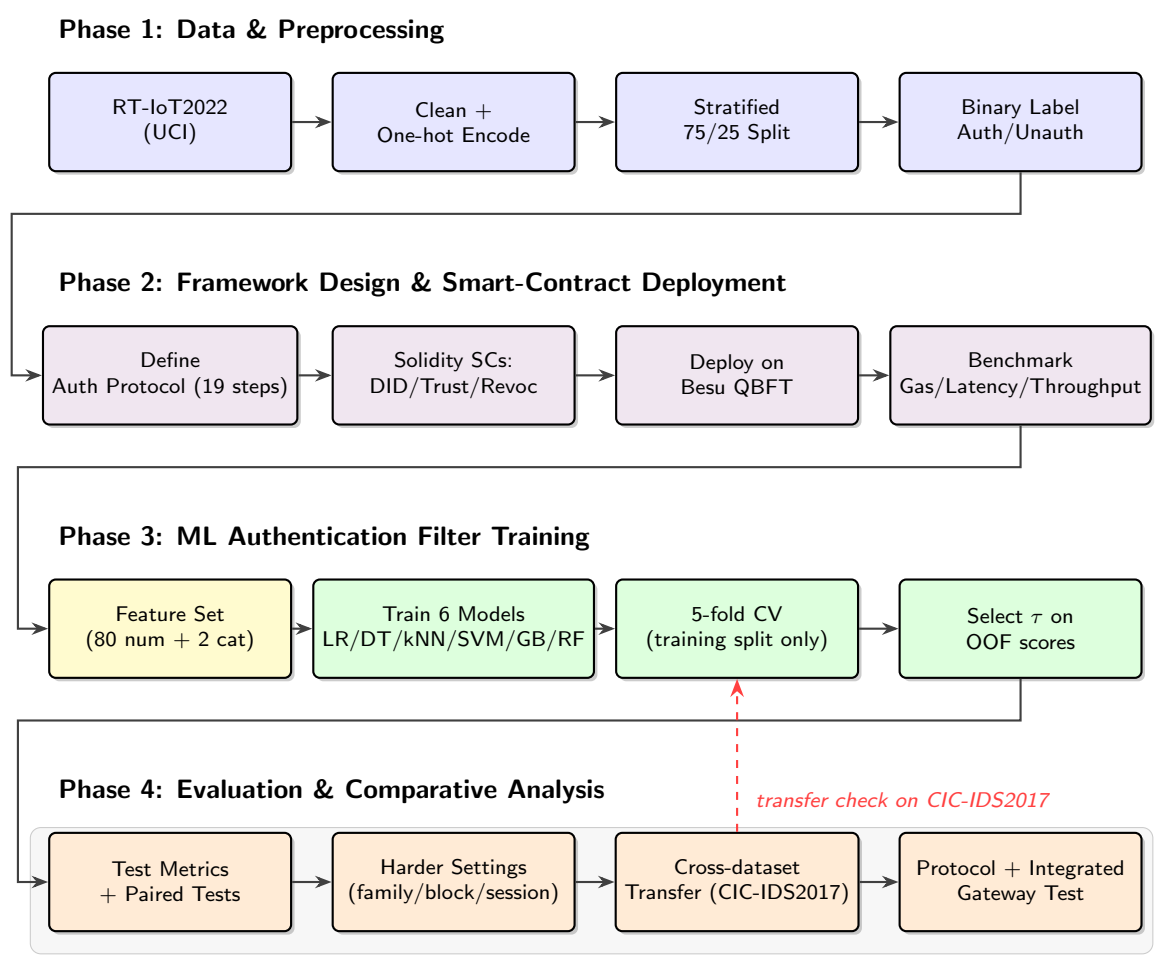
Phase 1: Data & Preprocessing
RT-IoT2022 (UCI)
Clean + One-hot Encode
Stratified 75/25 Split
Binary Label Auth/Unauth
Phase 2: Framework Design & Smart-Contract Deployment
Define Auth Protocol (19 steps)
Solidity SCs: DID/Trust/Revoc
Deploy on Besu QBFT
Benchmark Gas/Latency/Throughput
Phase 3: ML Authentication Filter Training
Feature Set (80 num + 2 cat)
Train 6 Models LR/DT/kNN/SVM/GB/RF
5-fold CV (training split only)
Select τ on OOF scores
transfer check on CIC-IDS2017
Phase 4: Evaluation & Comparative Analysis
Test Metrics + Paired Tests
Harder Settings (family/block/session)
Cross-dataset Transfer (CIC-IDS2017)
Protocol + Integrated Gateway Test